

Incorporating Spatial and Spatio-Temporal structures in Ecological models

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1 Introduction



Variability is all around

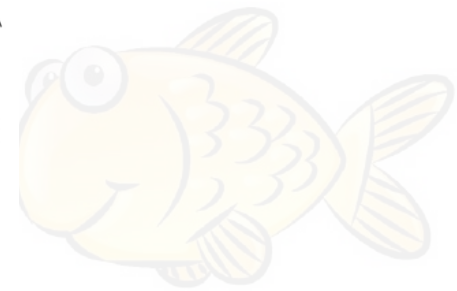
- Data obtained when performing any experiment show **variability**.
- **Statistics** is a discipline that has been developed in response to the experimenters whose data exhibit variability.
- How can we answer the question **where** things are, **in which amount** can we find them, and also, which is the **probability of finding** something in a particular location.
- In other terms, what if our **data are presented in terms of a map or georeferenced**?
- We will like to have models and methods that recognize the presence and importance of that **spatial information**.
- Indeed, **nearby observations tend to be more alike than those far apart**.

Spatial statistics (1)

- Tobler (1970) called this the first law of geography: **everything depends on everything else, but closer things more**.
- “A major pleasure in working with spatial data is their visualization. Maps are amongst the most compelling graphics, because the space they map is the space we think we live in, and **maps can show things we cannot see otherwise**” (Bivand, Pebesma, Gómez-Rubio, Applied Spatial Data Analysis with R, Springer, 2013).

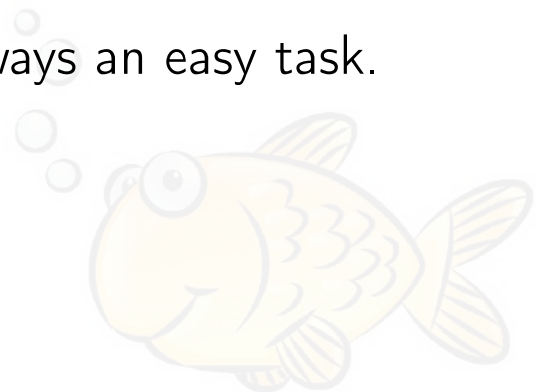
Spatial statistics (2)

Not a new thing: John Snow mapped out the spread of a cholera outbreak in London back in 1854 by relating with the location of pumps



Different types of spatial data imply different models

- Three prototypes:
 - 1 Geostatistical data
 - 2 Lattice/areal/regional data
 - 3 Spatial point patterns
- Distinguishing between them not always an easy task.

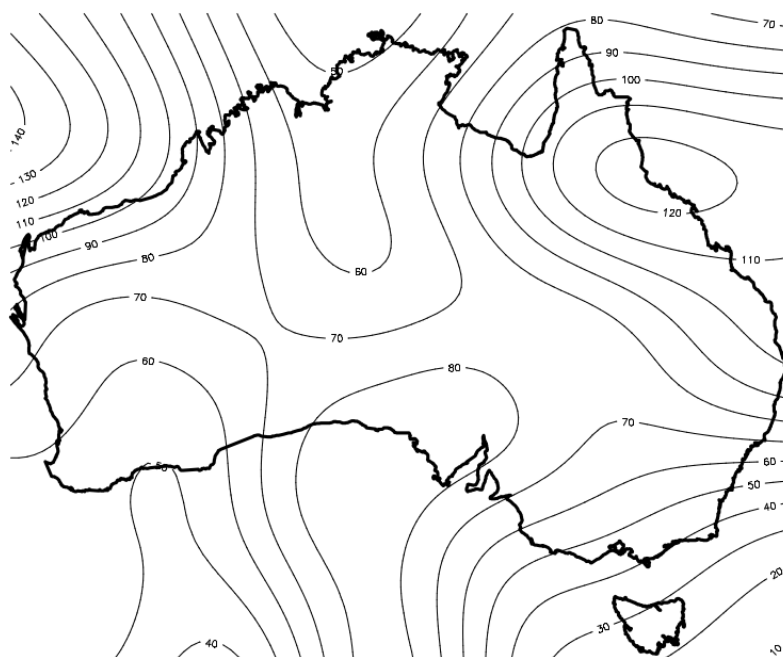


2 | Types of spatial data



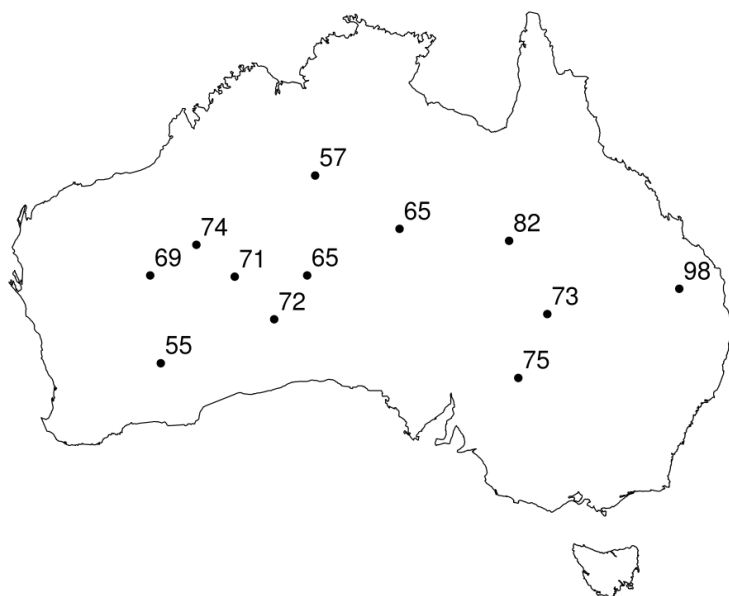
Geoestatistical data

The quantity of interest has a value at any location, ...



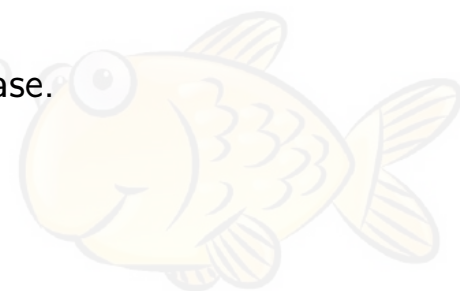
Geoestatistical data (2)

... but we only measure the quantity at certain sites. These values are our data.

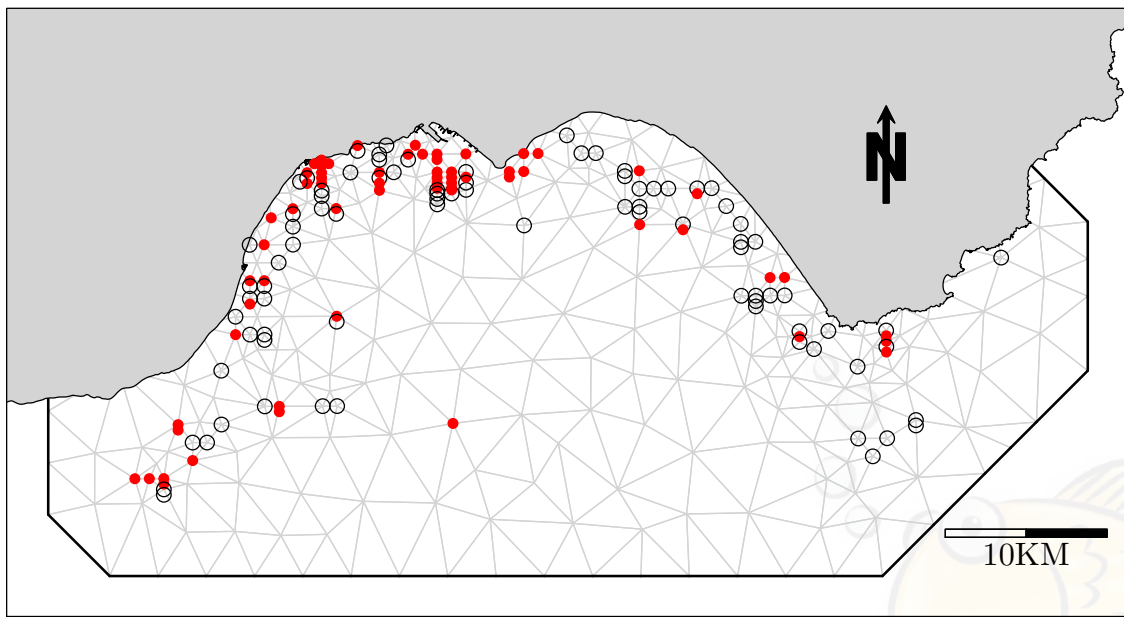


Geoestatistical data (3)

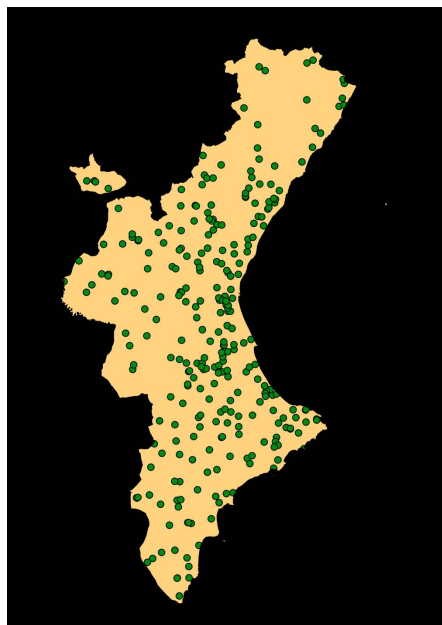
- In other words, point observations of a continuously varying quantity over a region.
- Final aim is to obtain (predict) the value of interest at any location.
- Examples:
 - ▶ A famous problem was to predict the ore grade in a mining block.
 - ▶ Weekly concentrations of ozone in Europe; annual acid rain deposition in Europe.
 - ▶ Abundance of a species; Prevalence of a disease.



Presence of Mackerel in Gulf of Almería

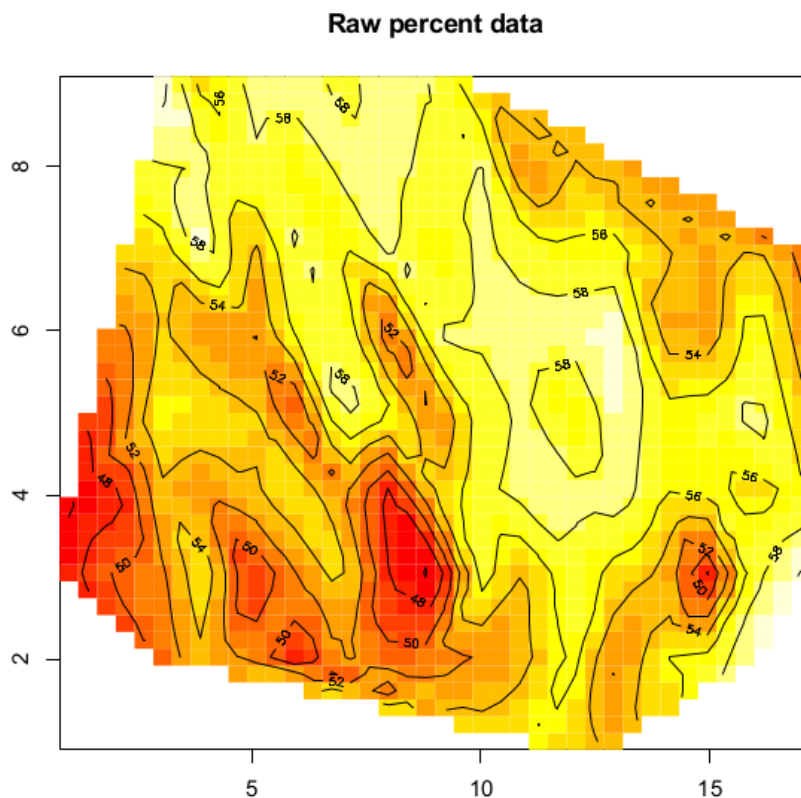
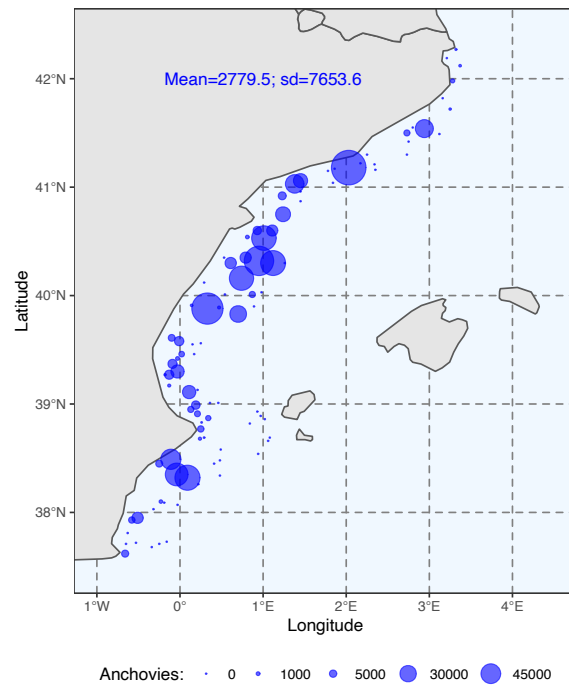


Weather stations



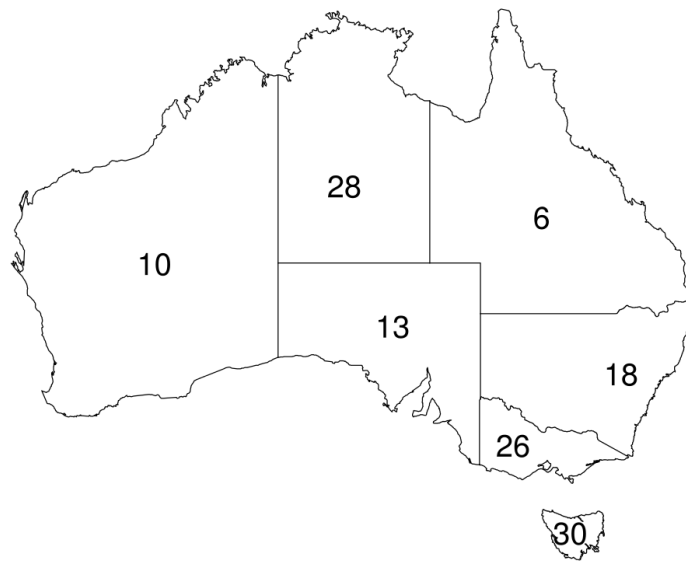
282 weather stations in Comunidad Valenciana: Daily records of Pluviometry, temperature, etc.

Fish species: captures of anchovies in the Mediterranean



Areal data

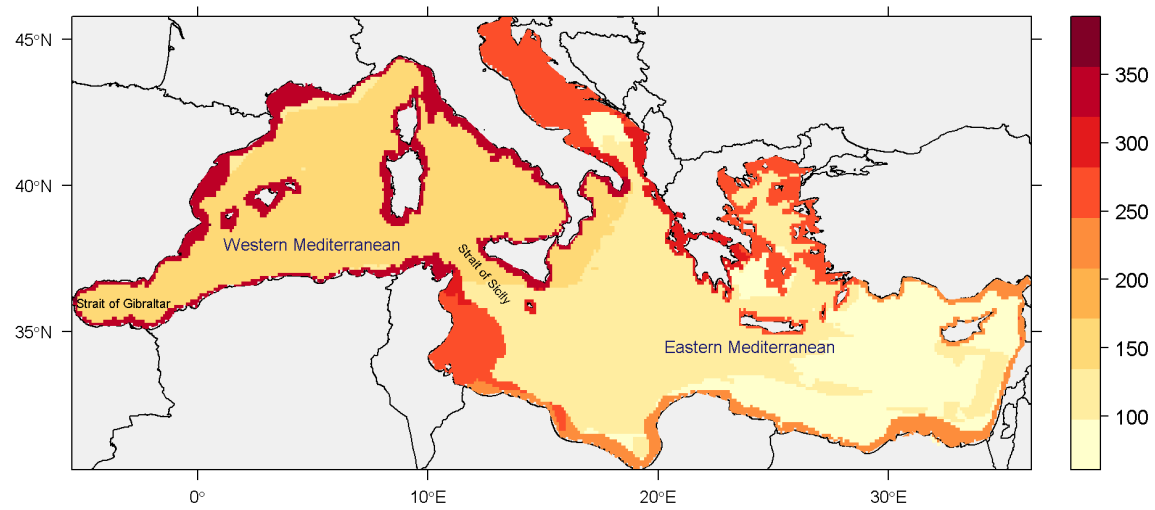
The quantity of interest is only defined for regions/municipalities/census districts It is measured/reported for certain fixed regions.



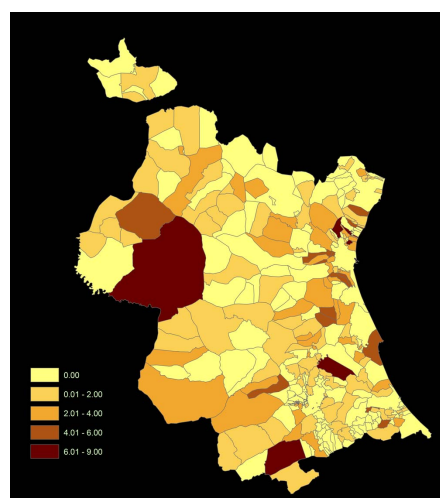
Areal data (2)

- Data are usually counts, averages or proportions of a quantity on subregions that make up a larger region.
- A lattice of locations evokes an idea of regularly spaced points. These will be referred to as regular lattices, allowing for the possibility of irregular lattices, whose relative displacements do not follow a predictable pattern.
- Statistical models for lattice data need to express the fact that observations nearby (in time or space) tend to be alike.
- Examples:
 - ▶ Presence or absence of a plant species in square quadrats over a study area
 - ▶ Number of deaths due to a disease in the provinces of a country
 - ▶ Number of different species in each quadrat of a regular grid

Species Richness in the Mediterranean Sea, Carmezim et al. (2022)

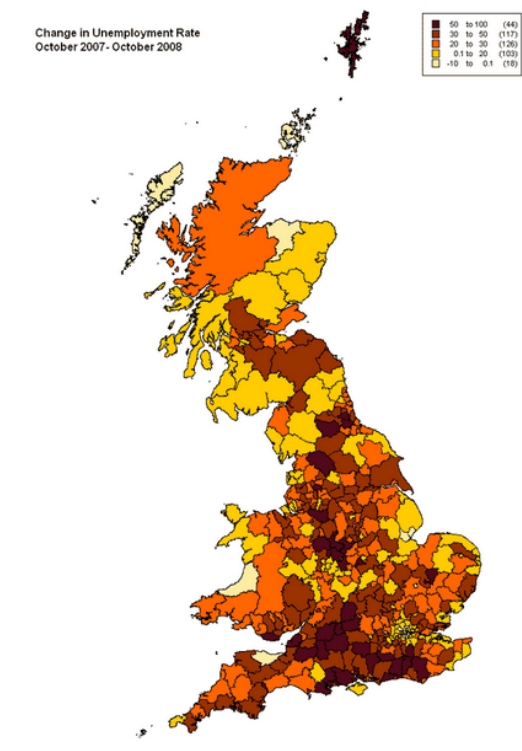


Disease mapping



Prostate cancer mortality (aggregated from 1975-1980). Colours indicate intensity of the characteristic studied.

Map of unemployment rates throughout the UK, from Oct. 2007 - Oct. 2008, as calculated by Local Futures



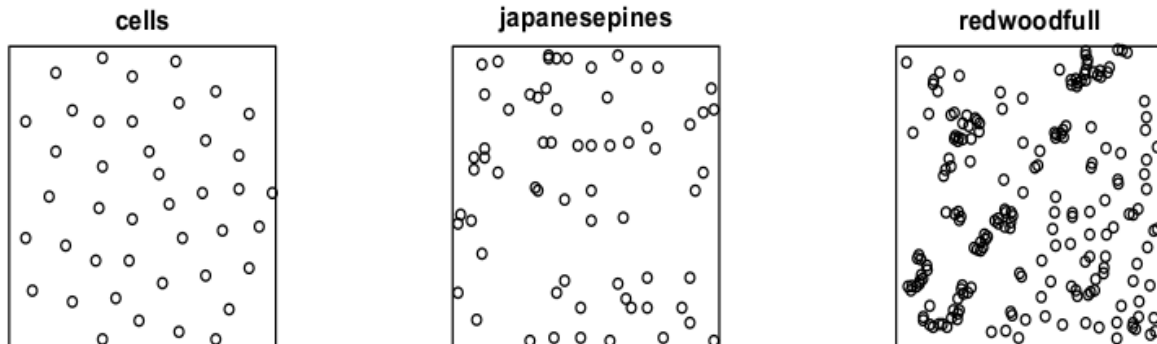
Point Pattern data

The main interest is in the locations of all occurrences of some event (e.g. tree deaths, meteorite impacts, robberies). Exact locations are recorded.



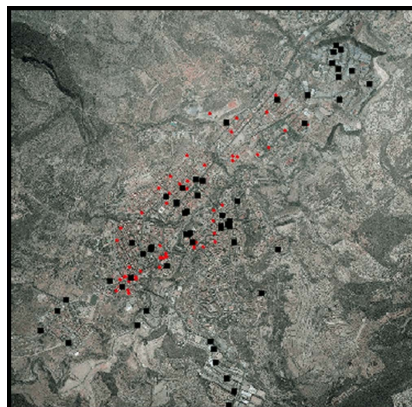
Examples of point pattern behavior

Differences in the maps show differences in the behaviour (different patterns)



Left: Regularity. Middle: complete spatial randomness. Right: clustering.

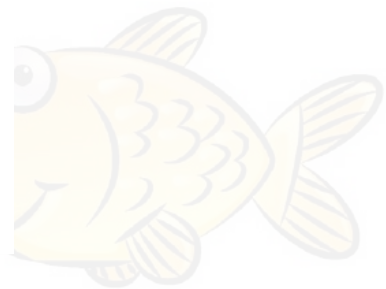
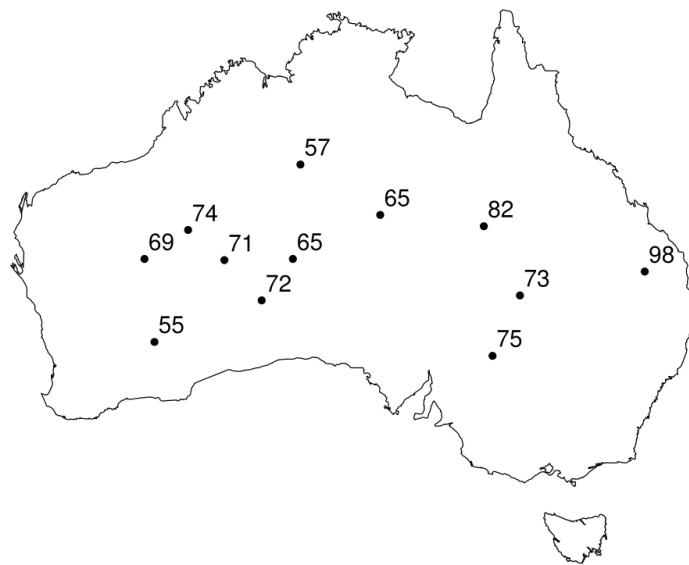
Legionella cases in Alcoy



The residence of affected people as a random location (red). Cooling towers as a possible cause (black)

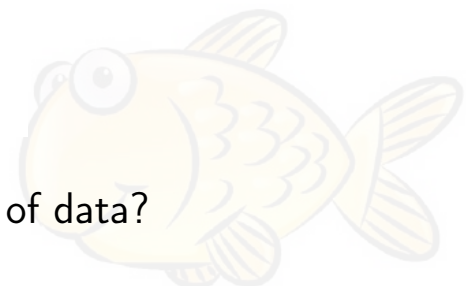
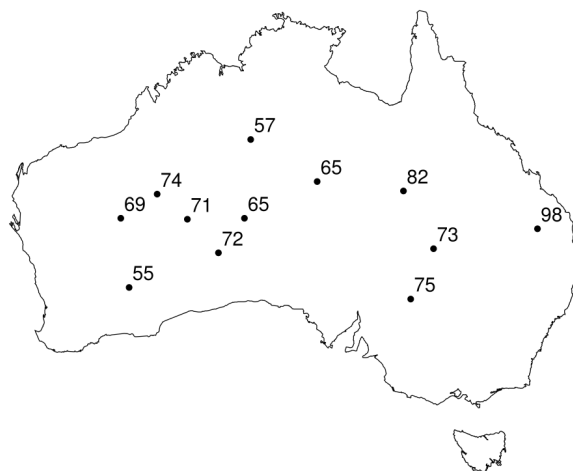
Points with marks

Points may also carry data (e.g. tree heights, meteorite composition, amount of the robbery)



Point pattern or geostatistical data?

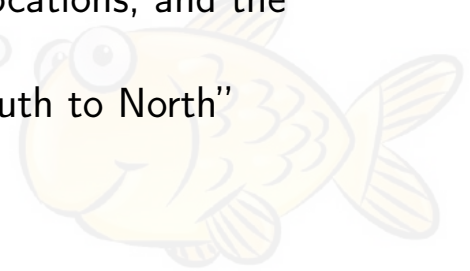
Graphs in pages 9 and 25 are (artificially) the same, ...



... but do they show the same type of data?

Point pattern or geostatistical data?

- **Response Variable** is the quantity that we want to be predicted or explained
- **Explanatory variable** is the quantity that can be used to predict or explain the response
- **Geostatistics** treats the spatial locations as explanatory variables and the values attached to them as response variables.
“Temperature is increasing as we move from South to North”
- **Spatial point pattern statistics** treats the spatial locations, and the values attached to them, as the response.
“Trees become less abundant as we move from South to North”



3 | Different models for different types of spatial data



How can we analyse spatial data?

- First of all: understanding the model behind each prototype of data.
- But the three prototypes can be seen under the same hierarchical structure.
- Hierarchical Statistical Modeling
 - ▶ Data model: distribution of the data given a hidden process
 - ▶ Process model: uncertainty in the true (but hidden) process through a probability distribution of the phenomenon of interest
 - ▶ Characterized by conditional probability distributions
- Once the model is set you need to find your way to make inference on the parameters involved: Bayesian.

Spatial models more detailed (Cressie, 2015)

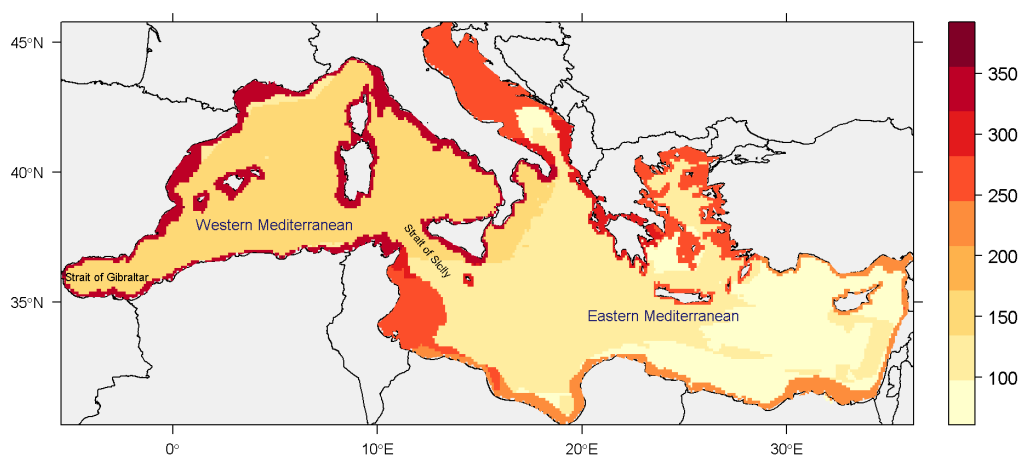
- Spatially continuous (Geostatistical or point-referenced) Data
 - ▶ R is a fixed subset of the plane of positive area (2-D) or volume (3-D).
 - ▶ $Y(s)$ is a random variable at each of the infinite continuous locations $s \in R$.
- Area (Lattice) Data
 - ▶ $R = \{s_1, s_2, \dots, s_n\}$ is a fixed regular or irregular lattice on the plane.
 - ▶ $Y(s_i)$ is a random variable at each of location s_i , $i = 1, \dots, n$.
- Spatial Point Process Data
 - ▶ $R = \{s_1, s_2, \dots, s_n\}$ is a random collection of points on the plane
 - ▶ $Y(s)$ is not specified or it is a random variable at a location $s \in R$ (marked point process).

4 Modelling and analysing areal/lattice data



Areal data

The **quantity of interest** is only defined for regions/quadrats It is measured/reported for certain fixed regions.



The way of modeling these **quantities only defined at certain fixed regions** is recognizing the fact that near quadrats should have similar information.

Hierarchical Bayesian Autocorrelated models

- Spatial Markovian assumption: value of the random variable at a given site only depends on the values at a specified set of neighboring sites.

- **Data Model**

$$Y(s_i) \sim \text{Distribution}(\theta_i)$$

where the distribution of response variable can be Normal (abundance), Poisson (number of species), Binomial, Beta (proportion of discards), etc.

- **Process model**

$$g(\theta_i) = \mathbf{X}(s_i)\boldsymbol{\beta} + Z_i$$

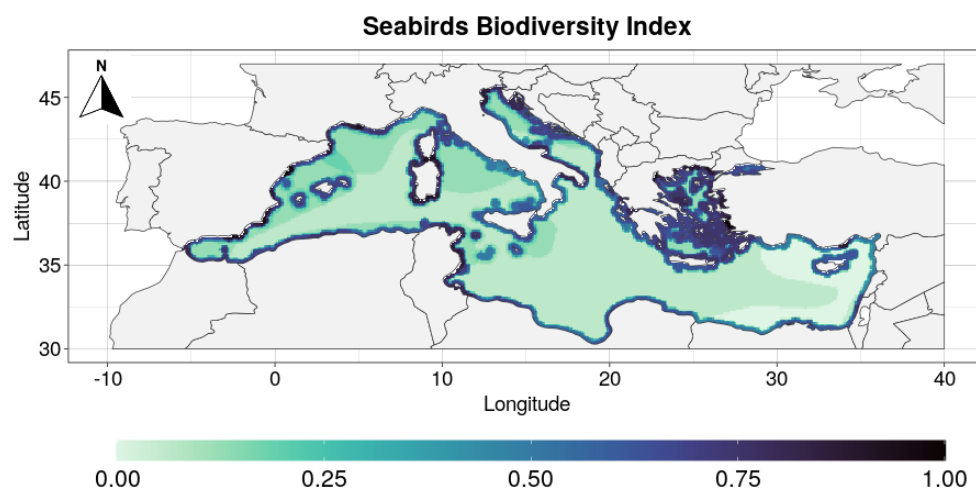
- ▶ $\mathbf{X}(s_i)$ known covariates
- ▶ $\boldsymbol{\beta}$ fixed effects parameters
- ▶ Spatial random effect has an Intrinsic Conditional Auto-Regressive distr.

$$[Z_i | Z_j, i \neq j, \tau_Z^2] \sim N \left(\bar{Z}_i = \frac{1}{\sum_j w_{ij}} \sum_j Z_j w_{ij}, \tau_i^2 = \frac{\tau_u^2}{\sum_j w_{ij}} \right)$$

where $w_{ij} = 1$ if adjacent (0 otherwise).

- ▶ Predictor can incorporate smooth nonlinear effects of covariates, time trends, seasonal effects, random intercept and slopes and temporal random effects.
- The prior for the hyperparameter τ_u must be assigned.

Analyzing seabirds biodiversity index (Sarzo et al., 2023)



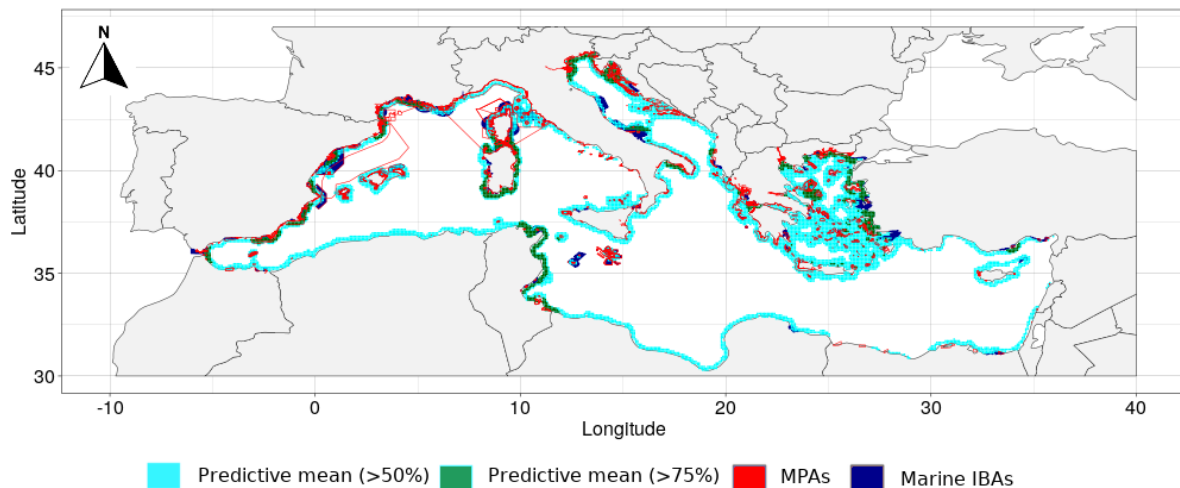
Seabirds biodiversity index for the Mediterranean Sea

Analyzing seabirds biodiversity index (Sarzo et al., 2023) (2)

As biodiversity index is scaled in the unit interval (0,1), a Bayesian Spatial Beta regression model is needed to analyze it:

$$\begin{aligned}Y_i &\sim \text{Beta}(\mu_i, \phi), \\ \text{logit}(\mu) &= \mathbf{X}_i \boldsymbol{\beta} + u_i, \\ \phi &= \exp(\theta), \\ \beta_j &\sim N(0, \tau = 10^{-3}), \\ u_i | u_j, i \neq j, \tau &\sim N\left(\frac{1}{\mathcal{N}_i} \sum_{i \sim j} u_j, \frac{1}{\mathcal{N}_i \tau}\right), \\ \tau &\sim \text{PC-prior}(5, 0, 1), \\ \theta &\sim \text{LogGamma}(1, 0, 01).\end{aligned}$$

Analyzing seabirds biodiversity index (Sarzo et al., 2023) (3)



Overlap among posterior predictive mean values over 0.50 (cyan), 0.75 (green), MPAs (red), and marine IBAs (dark blue) in the Mediterranean Sea.

- Disease mapping is probably the most typical example of areal data.
- The main idea is to assess the spatial pattern of a particular disease and to identify areas characterized by unusually high or low relative risk.
- Data are discrete: they represent the counts of people with the disease or deaths in each area analyzed.
- Data by themselves are not comparable: it is not the same 100 deaths in a large city than in a small town.
- These counts are usually (mostly in the epidemiological context) transformed in standardized mortality ratios (SMR), which are defined as the ratio between the number of observed cases y_i and the number of expected cases E_i in each area i .
- To understand the truly process underneath these SMRs, we have to model the spatial dependence among the areas.
- In seminal works in the 70s, Besag et al. presented a way to analyze areal data taking into account the spatial dependencies.

The Besag-York-Mollié approach

- As data are counts, we use a Poisson distribution. For each area i ,

$$y_i \sim \text{Poisson}(\lambda_i), \quad \lambda_i = E_i \rho_i, \quad \log(\rho_i) = \eta_i,$$

where λ_i is defined in terms of a rate ρ_i and of the expected number of cases E_i .

- Apart from the spatially structured random effect (and any possible covariates), Besag et al. proposed also the use of a non-structured random effect in the linear predictor η_i :

$$\eta_i = \mathbf{X}(s_i)\beta + u_i + v_i,$$

where $\mathbf{u}$ is the independent random effect and $\mathbf{v}$ is the spatially structured random effect.

- The Intrinsic Conditional Auto-Regressive (ICAR) distribution for the spatial effect:

$$u_i \sim N(0, \tau_{\mathbf{u}}^{-1}), \quad v_i | \mathbf{v}_{-i} \sim N\left(\frac{1}{n_i} \sum_{j \sim i} v_j, \frac{1}{n_i \tau_{\mathbf{v}}}\right).$$

- The priors for the hyperparameters $\tau_{\mathbf{u}}$ and $\tau_{\mathbf{v}}$ are usually expressed with a reparametrization in terms of their logarithms:

$$\log(\tau_{\mathbf{v}}) \sim \log\text{Gamma}(1, 5 \cdot 10^{-5}), \quad \log(\tau_{\mathbf{u}}) \sim \log\text{Gamma}(1, 5 \cdot 10^{-5}).$$

- Note that this is latent Gaussian (field) model, in fact, a Gaussian Random Markov field and so, it is perfectly suited to be analyze within INLA (Rue et al., 2009).

Example of disease mapping: Larynx cancer

Larynx cancer mortality counts are observed in the 544 districts of Germany from 1986 to 1990. Together with the counts, for each district, the level of smoking consumption c is registered. The Besag-York-Mollie model is given by:

- 1 Data are conditionally independent Poisson r.v.s with mean $E_i \exp(\eta_i)$, where E_i is fixed and accounts for demographic variation, and η_i is the log-relative risk:

$$y_i | \eta_i \sim \text{Poisson}(E_i \exp(\eta_i)), \quad i = 1, \dots, 544$$

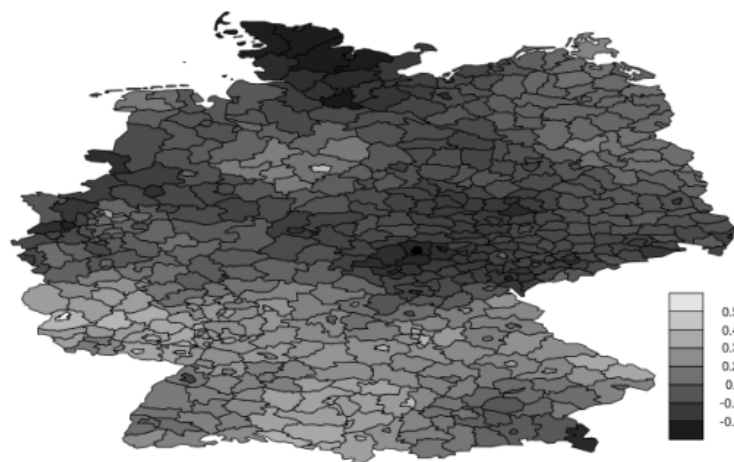
- 2 The linear predictor η_i is formed by an intercept μ , a smooth function of the smoking consumption covariate, a spatial model over different district locations, and an unstructured random effect to account for overdispersion:

$$\eta_i = \mu + f_s(s_i) + f(c_i) + u_i$$

where $\mathbf{x} | \boldsymbol{\theta}$ is Gaussian distributed.

- 3 Assigning priors for $\boldsymbol{\theta}$, formed by the precision parameters associated with the smooth model for the covariate $f(c_i)$, the spatial model $f_s(s_i)$ and the random effect u_i .

Example of disease mapping: Larynx cancer (2)



Posterior mean of the spatial effect in each of the 544 districts in Germany.

5 | Modelling and analysing Geostatistical data



Bayesian Hierarchical Geostatistical models.

- Different response variables (normal, Poisson, binomial, beta, etc.) depending of the situation.
- **Data Model** $\mathbf{y} = (y_1, \dots, y_n)$ represents the observed values of the corresponding **variable of interest** with mean $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)$ at n locations.
- **Process model** each μ_i can be easily linked to a **structured additive predictor** η_i and a **geostatistical spatial random effect**:

$$g(\mu_i) = X_i\beta + W_i, i = 1, \dots, n.$$

The **spatial effect** W is Gaussian: it is a **Latent Gaussian field**.

- This predictor can incorporate also smooth nonlinear effects of covariates, time trends and seasonal effects, random intercept and slopes as well as temporal random effects.
- **Prior distributions** for the parameters and hyperparameters need to be assigned.

- But geostatistics involves working with **continuously indexed Gaussian Fields** and INLA cannot be applied directly.
- Lindgren et al. (2011) proposed an explicit link between *Gaussian fields* and *Gaussian Markov random fields* (GMRF): the **Stochastic Partial Differential Equation approach (SPDE)**.
- GMRFs are discretely indexed → the Markov property makes the precision matrix involved sparse (allowing the use of faster numerical algorithms).

Final modelling using SPDE

$$\begin{aligned}
 Z_i &\sim \text{Distribution}(\theta_i), i = 1, \dots, n \\
 g(\theta_i) &= X_i\beta + w_i \\
 \pi(\beta_j) &\sim N(\mu_{\beta_j}, p_{\beta_j}) \\
 \mathbf{w} &\sim N(0, \mathbf{Q}(\kappa, \tau)) \\
 2\log\kappa &\sim N(\mu_{\kappa}, p_{\kappa}) \\
 \log\tau &\sim N(\mu_{\tau}, p_{\tau})
 \end{aligned}$$

- The spatial effect depends on now two different parameters: κ (the range of the effect) and τ (the total variance). Prior information for them via pc-priors (Simpson et al., 2017).
- Once the inference is performed, the model is used to predict in unsampled places (providing a map of the species behaviour).

5.1 Modelling occurrences



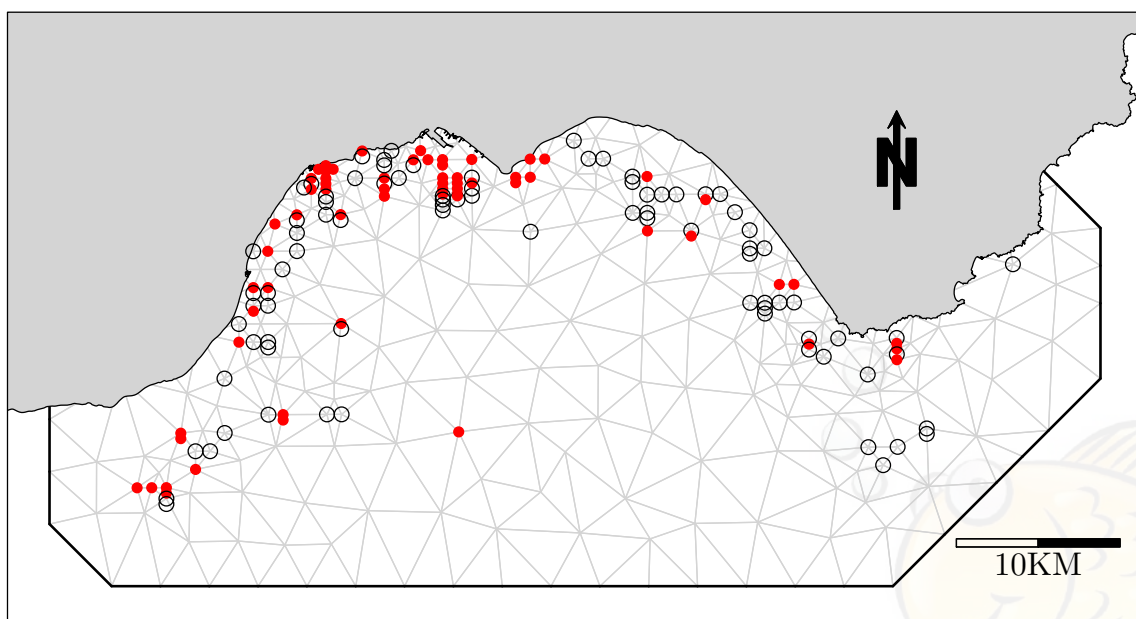
Logistic regression models for presences and absences

When the **observed variable** is the **ocurrence** (presence/absence) of a particular characteristic, we obtain **maps of the posterior predictive distribution of the probability of finding the characteristic** (species, disease, pollutant, etc.):

- Distribution of Mediterranean horse **mackerel** in Gulf of Almería (Muñoz et al., 2013).
- Distribution of prevalence of **Bovine paramphistomosis** in Galicia (González-Warleta et al., 2013).
- Distribution of prevalence of **Citrus Black Spot** (Martínez-Minaya et al., 2018).
- Distribution of vulnerable (to fishing pressure) **elasmobranch** species (Pennino et al., 2013).
- Distribution of prevalence of **Xylella fastidiosa** in Alicante (Cendoya et al., 2020).

Occurrence of Mackerel in Gulf of Almería

The measures are presences (**red**) and absences (**white**)



To model the distribution of the presence or absence of mackerel, we define a **binary random variable** as a response variable:

$$Z_i \sim \text{Ber}(\pi_i),$$

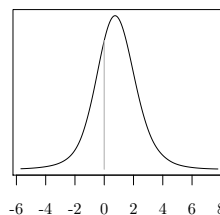
(presence (1) or absence (0) of the species).

- **Probability of occurrence**

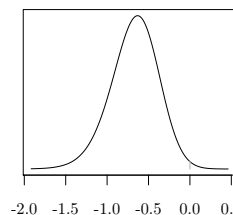
$$\text{logit}(\pi_i) = X_i\beta + W_i \quad i = 1, \dots, n$$

- **Linear predictor** for observation i
- **Spatial effect**
- We consider **prior Gaussian distributions** for all the fixed effects in β and **pc-priors** for the parameters of the spatial effect.

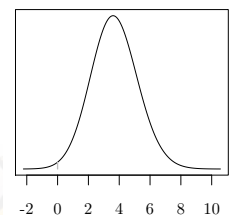
- In spite of its low commercial value, mackerel plays an important role in the observed transition zone between the Mediterranean and Atlantic sea.
- **Covariates**: geographical coordinates, bathymetry, monthly precipitation, sea surface temperature, chlorophyll-a concentration.
- We used DIC and CPO to **compare models** with **different combinations** of the available covariates.
- Final model includes **log(Depth)** (negative effect) and **chlorophyll-a** (positive effect).



Intercept

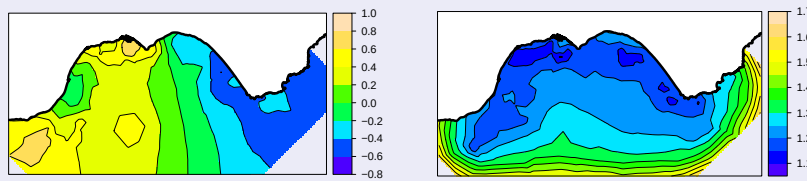


log(Depth)



chlorophyll-a

Spatial effect

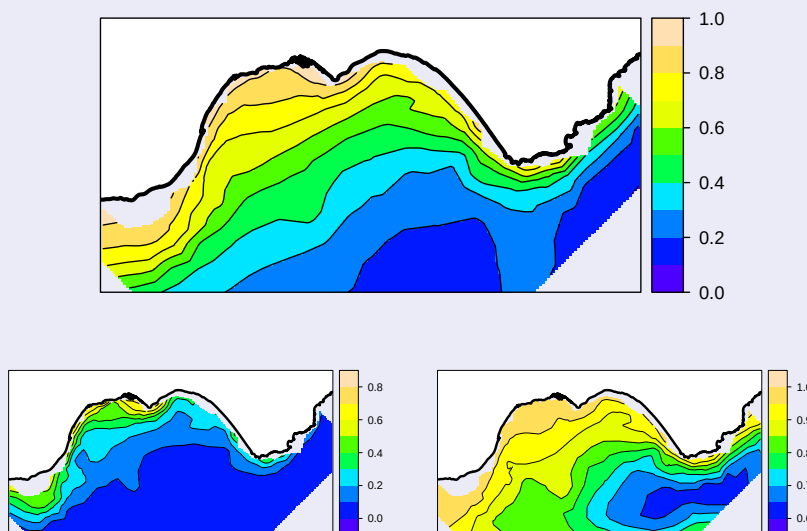


Posterior mean and standard deviation of the spatial effect

Interpretation

- There seems to be a **east-west effect**.
- This was associated a posteriori by local experts with the fact that **the western area of the bay is a protected coastline** with favourable conditions for the species.
- The model provided (unexpectedly) a **quantification of the impact of this protective action** on the Mediterranean horse mackerel.

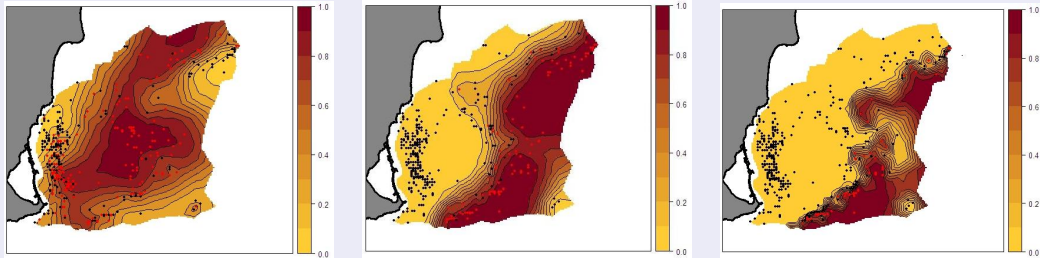
Posterior probability of occurrence



Median (top) and quartiles 1 and 3 of $\pi_i|\text{data}$

- There is an increasing concern over **elasmobranch species** because they are **highly vulnerable to fishing** pressure.
- **Main predictors of elasmobranch habitats** are depth, slope of seabed and type of substrate, followed by temperature and chlorophyll-a.

Median of the posterior probability of the presence of elasmobranch species



- Species show **different optimum depths**: could indicate a **sort of fine-tuned bathymetric segregation**, though they coexist on shelf and slope bottoms.
- These maps can be used to **identify sensitive habitats**, with a final aim to improve management and conservation of these vulnerable species.

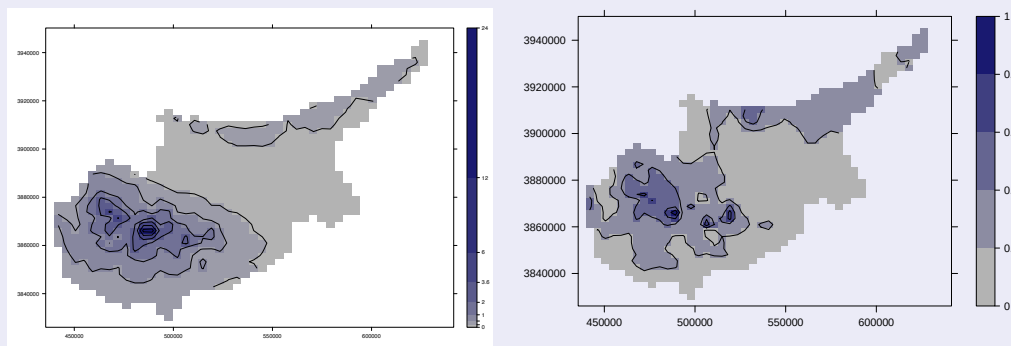
5.2 | Modelling positive continuous quantities

Dealing with positive continuous distributions

The **observed variable** can be also abundances (normal, gamma, lognormal, etc.):

- **Abundance of hake recruitment** (Paradinas et al., 2015, 2017; Izquierdo et al., 2021), **amount of discards** (Pennino et al., 2014), **abundance of red mullet** (Paradinas et al., 2020), **CPUE biomass indices** (Fuster-Alonso et al. 2024).
- **Milk protein content** of cows with fasciolosis (González-Warleta et al., 2024).
- Spatial distribution of **bioclimatic indices** (Barber et al., 2017).

Mean (left) and standard deviation (right) of the posterior predictive distribution of the Ombrothermic Index



Modelling the amount of pollutants.

- The corresponding model for the **level of a pollutant** (for instance PM2.5 concentration level) using a continuous (**Gaussian**) distributed response:

$$Y_i \sim N(\mu_i, \sigma^2), i = 1, \dots, n.$$

- The **mean of the level of pollutant** μ_i is **linked with the effect of covariates** (e.g. temperature and precipitation) and the **spatial random effect**:

$$\mu_i = \beta_0 + \beta_1 \text{temp}_i + \beta_2 \text{prec}_i + W_i, i = 1, \dots, n$$

with $\mathbf{W} \sim N(0, \mathbf{Q}(\kappa, \tau))$ a Latent Gaussian field with $\mathbf{Q}(\kappa, \tau)$ determined by the Matérn covariance.

- Priors for parameters ($\beta_0, \beta_1, \beta_2 \sim N(0, 10^{-6})$) and hyperparameters ($1/\sigma^2 \sim \text{Gamma}(1, 0.0001)$, $2\log\kappa \sim N(\mu_\kappa, p_\kappa)$ and $\log\tau \sim N(\mu_\tau, p_\tau)$).

5.3 Dealing with proportions

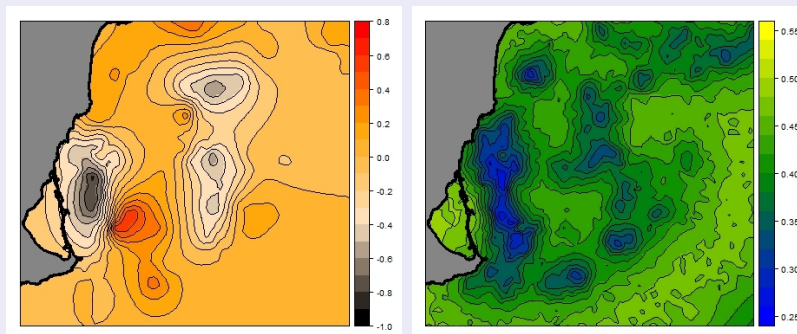


Beta spatial regression for proportions

The **observed variable** can also be expressed in terms of proportions:

- Paradinas et al. (2016, 2018) have applied spatial Beta regression models to analyse the **proportion of discards**, and **identified two main fishing-suitable areas** based on the proportion of discarded regulated fish proportions.

Posterior mean (left) standard deviation (right) of the spatial component of the regulated species discard proportion



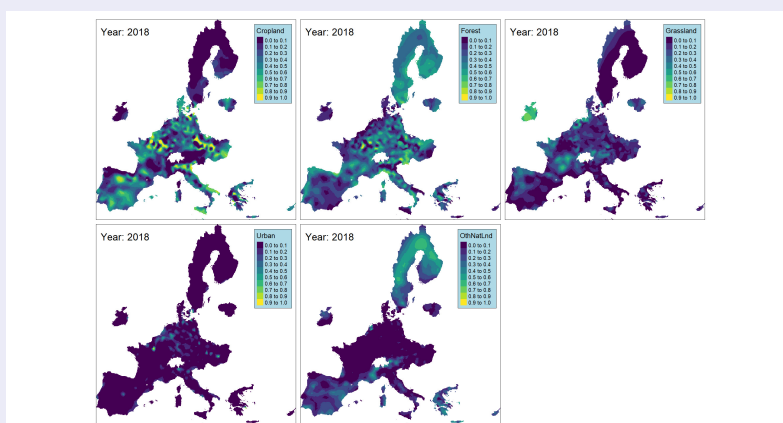
- Martínez-Minaya et al. (2019) have analysed the **proportion of arapbidosis thaliana** to study the **effect of climate change**.
- Castilla et al. (2020) have also analysed the ecological, genetic and evolutionary drivers of regional genetic differentiation in *Arabidopsis thaliana*.

Dirichlet regression for vectors of proportions

The **observed variable** can be expressed in terms of proportions of different groups:

- **Compositional data** $\mathbf{Y} = \{\mathbf{y}_{c \times i}\}$ are defined as a vector of values in the interval $(0,1)$ that satisfies $\sum_{c=1}^C y_{ci} = 1 : c \in \{1, \dots, C\}$ and $n \in \{1, \dots, N\}$.
- In other words, data consisting in **proportions** (of categories) **adding up to 1**.
- Martínez-Minaya and Rue (2024) have re-analysed the **proportions** of *Arapdidosis Taliana* in Peninsula Iberica to study the **effect of climate change**.
- Figueira et al (2025) have analysed **proportions of land-use in Europe**.

Posterior predictive distribution of proportions of land use.



5.4 Dealing with counts

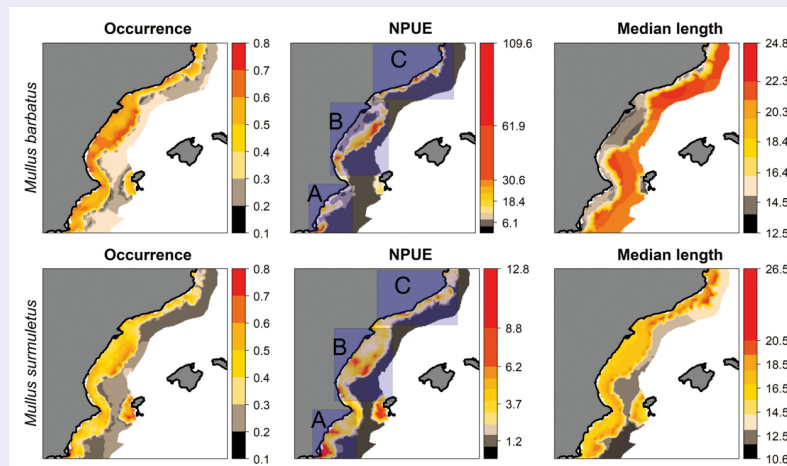


Regression models for counts

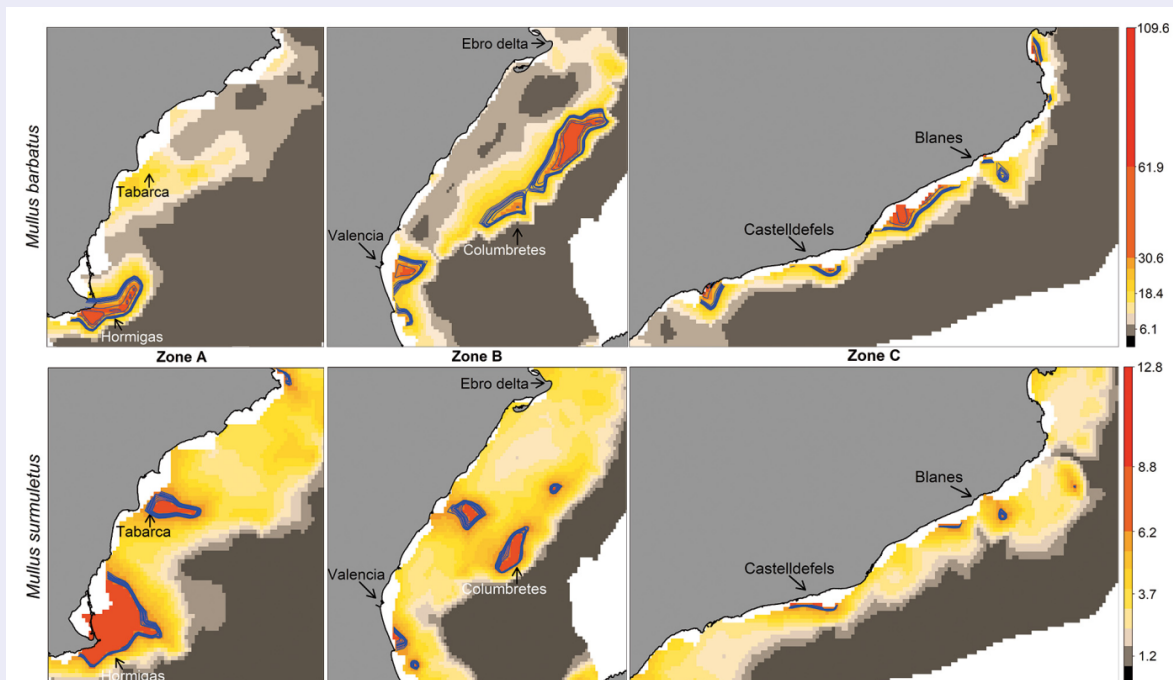
The **observed variable** can also be expressed in terms of counts:

- Paradinas et al. (2020) have considered spatial Negative Binomial regression models to analyse the **number of catches per 30 minutes** (NPUE) in order to understand the spatiotemporal persistence of fish distributions and so help to define fish hotspots and effective fisheries-restricted areas.

Average spatial distributions of *Mullus barbatus* and *M. surmuletus* occurrence, NPUE and median length (cm) between 2000 and 2016. Blue rectangles represent zoomed sections in next slide.



Zoomed spatial distribution of *Mullus barbatus* and *M. surmuletus* NPUE in Murcia (Zone A), Valencia (Zone B) and Catalonia (Zone C). Hotspots, identified as areas with NPUE above the 90 th percentile, are highlighted in blue



6 | Analyzing Point Patterns with Point Processes



Spatial point patterns

- Data are presented as **locations of occurrence of an event**.



- These data can also come with a mark (a random variable expressing a value of interest in that location).
- Interest is to analyse the **geometrical structure of the pattern** (complete spatial randomness, clustering, or regularity) formed by those data in the space.
- But, the tendency of the human eye to see clustering or other structure in virtually every point pattern renders a strictly graphical approach unreliable.

Spatial point processes

- A **point pattern** is a **collection of points** in some area or set and is typically interpreted as a sample from (or **realisation** of) a **point process** (a particular case of stochastic process).
- A homogenous (stationary) Poisson point process $pPo(\lambda)$ is a point process with the following properties
 - ▶ It is **Stationary**, invariant through traslation;
 - ▶ It is **Isotropic**, invariant through rotation;
 - ▶ λ is called **intensity of the process**, it is constant in all the space analysed and it can be interpreted as the mean number of points per unit area; or ...
 - ▶ ... in other words, the **number of points of the process in any bounded set B follows a Poisson distribution** with mean $\lambda \times \text{Area of } B$;
 - ▶ It has the **Completely random property**, the intensity is constant and each process is independent of all the rest.
- Poisson point processes produce point patterns (realisations) of **complete spatial randomness** (the graph in the middle in previous slide). **Deviations from that model because of other possible effects should then be analysed.**

Inhomogeneous Poisson Point Processes

- An **inhomogeneous Poisson process** (also named Cox Process) is a straightforward generalisation of the homogeneous Poisson process when there is no stationarity.
- In other words, the constant intensity of the homogeneous Poisson process λ is now replaced by an **intensity function $\lambda(x)$ whose value varies with the location x .**
- The **number of points in any bounded set B has a Poisson distribution** with mean $\int_B \lambda(x) dx$.
- When the **log intensity of the Cox process is modeled with a Gaussian linear predictor**, the Cox process is known as a **log-Gaussian Cox process**:

$$\log(\lambda(x)) = \mathbf{X}\beta + \lambda_0(x),$$

where $\lambda_0(x)$ is an underlying spatial distribution modulated by the vector of covariates.

- $\lambda_0(x)$ can be estimated using kernel smoothing methods, or its log can be considered as a **Gaussian Field** with Matérn covariance as in the Geostatistical framework.

Hierarchical Bayesian modeling of LGCPs

- Data Model: data Y_i are a realization of an inhomogeneous Poisson point process with intensity λ_i .
- Process model:

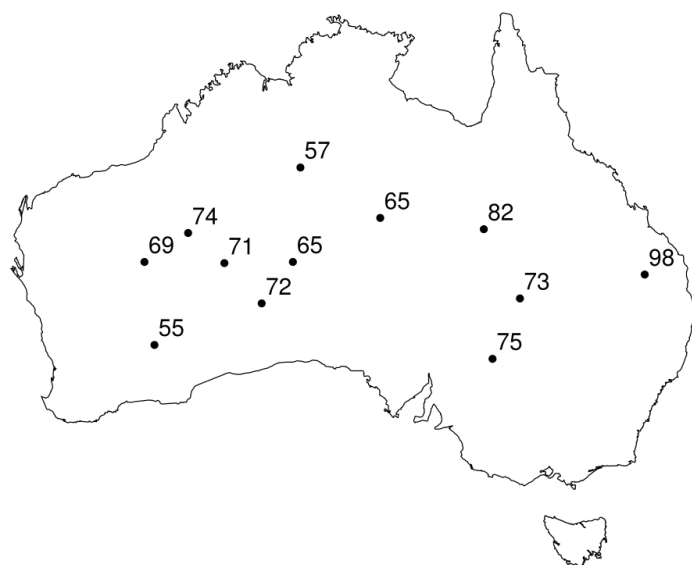
$$\log(\lambda_i) = \mathbf{X}_i\boldsymbol{\beta} + W_i, i = 1, \dots, n$$

with:

- ▶ $\mathbf{X}_i$ known covariates at each location i ,
- ▶ $\boldsymbol{\beta}$ fixed effects parameters
- ▶ $\mathbf{W}$ is a stationary and isotropic Gaussian spatial process (GF) with a Matérn covariance with parameters (κ, τ) .
- Other effects can also be incorporated.
- Priors are needed for the parameters and hyperparameters.
- If the covariates and the spatial effect are not relevant, the generating process is a Homogeneous Poisson process with intensity constant λ .

Marked point processes

- Points may also carry data (e.g. tree heights, meteorite composition, amount of the robbery)



- Model for the Marks

- ▶ Data Model

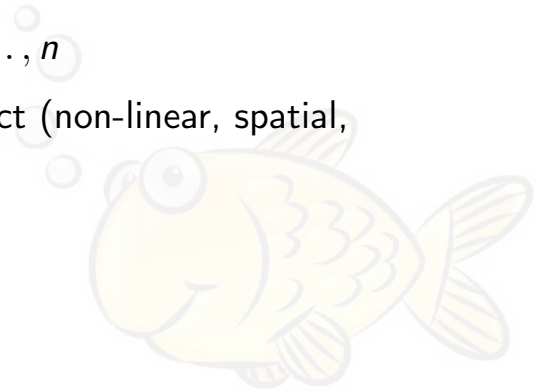
$$Y(s_i) \sim \text{Distribution}(\theta_i)$$

where the distribution of response variable can be Normal, Poisson, Binomial, etc.

- ▶ Process model

$$g(\mu_i) = X_i\beta, i = 1, \dots, n$$

This predictor can incorporate any other effect (non-linear, spatial, temporal, etc.).



Hierarchical Bayesian modeling of Marked Point patterns (2)

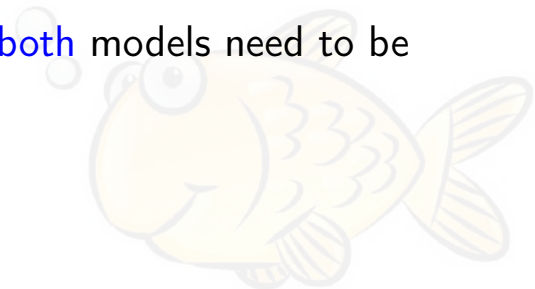
- Model for the locations is a LGCP

- ▶ Data model, Y_i are a realisation of an inhomogeneous Poisson point process with intensity λ_i ;
 - ▶ Process model,

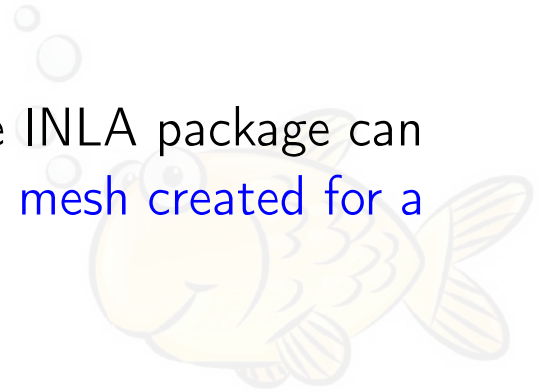
$$\log(\lambda_i) = \mathbf{X}_i\beta + W_i, i = 1, \dots, n$$

with $\mathbf{W}$ a stationary and isotropic Gaussian spatial process (GF) with a Matérn covariance with parameters (κ, τ) .

- Priors for parameters and hyperparameters in both models need to be assigned.



- The spatial process $\mathbf{W}$ can be estimated using the [SPDE approach](#) described in the Geostatistical Data section.
- The [inlabru](#) package provides a simple interface for the analysis of point patterns with INLA and SPDE models.
- The [meshbuilder\(\)](#) function in the INLA package can also help to [test the adequacy of the mesh created for a SPDE latent effect](#).



7 | Geostatistical models with preferential sampling



Dealing with preferential sampling.

- Geoestatistical models usually assume that sampling locations and the process being modelled are stochastically independent.
- But, sometimes sampling locations are **deliberately concentrated** in **areas where the abundance of the variable of interest** is known or expected to be high:
 - ▶ fishermen **fish in areas where they are likely to find fish**,
 - ▶ in mineral exploration, samples may be concentrated in areas that are thought likely to yield high grade ore;
 - ▶ measurements of pollution are usually taken where more pollutants are (cities, etc.).
- In this case, the **quantity observed can be considered as a mark** that depends on the latent Gaussian Field which creates the realisations of the LGCPs.
- Instead of a geostatistical model, we have a **marked point pattern** that it is a **realisation of a Marked point process** with two sub-models written in terms of GFs.

Model for the marks

- **Model for the quantities observed at each location**, $\mathbf{y} = (y(s_1), \dots, y(s_n))$, are assumed to follow an exponential family distribution whose mean is related with the spatial term:

$$\begin{aligned} Y(s_i) &\sim \text{Distribution}(\theta_i) \\ g(\mu_i) &= \mathbf{X}_i \boldsymbol{\beta} + W_i, \quad i = 1, \dots, n. \end{aligned}$$

being $\mathbf{W}$ the underlying (latent) continuously-indexed (multivariate) Gaussian Field (GF) with covariance with parameters (κ, τ) .

- The **sampling design process** (point pattern) **depends on this unknown target $\mathbf{W}$ spatial stationary Gaussian process**.

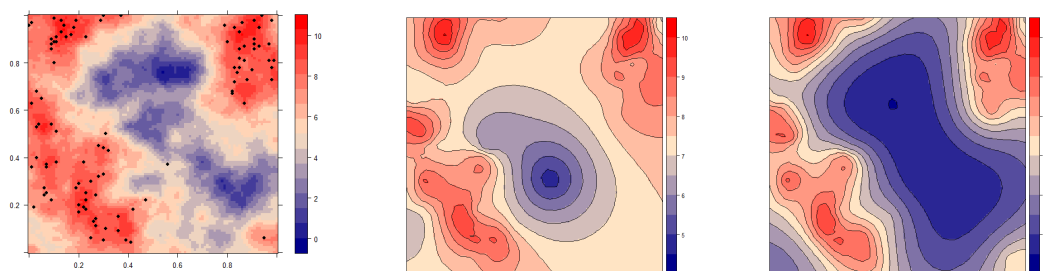
Model for the locations

- The **observed locations** $(s_1, \dots, s_n)$ are a **realisation of an inhomogeneous Poisson point process** with intensity λ_i ;

$$\log(\lambda_i) = \mathbf{X}_i^{(*)} \boldsymbol{\beta}^{(*)} + \alpha W_i, i = 1, \dots, n$$

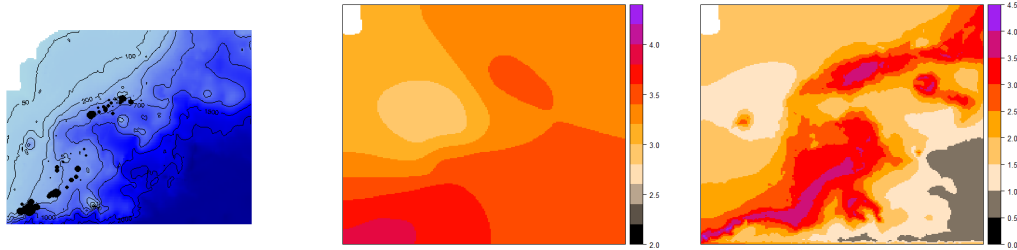
- ▶ with **$\mathbf{W}$** the previous **stationary and isotropic Gaussian spatial process** (GF) with a Matérn covariance with parameters (κ, τ) ,
- ▶ and α is the **shared parameter** between both likelihoods that express the preferential sampling.
- The parameter α **expresses the preferential sampling**: values of α close to 0 would indicate no preferential sampling, while values larger than 0 would indicate preferentiality and values below 0 would indicate negative preferentiality.
- Priors for parameters and hyperparameters in both models need to be assigned.

Simulated example



- Left: **Simulated Gaussian field and point pattern** representing the **sampling locations**.
- Middle: Posterior predictive mean maps of the simulated abundance process **without preferential sampling** correction ...
- Right: ... and **with the preferential sampling correction**.

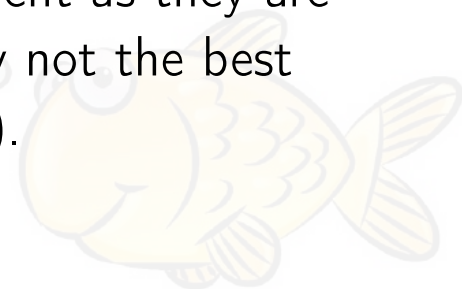
Red shrimp near Alicante: the preferentiality appears in the covariate shared



- Left: Abundance data at the sampling locations,
- Middle: posterior predictive mean of the red shrimp distribution without the preferential bathymetric correction ...
- Right: ... and with the preferential bathymetric correction,
- the non-preferential model prediction seems driven by the spatial effect while the preferential model map corrects the bathymetric effect providing a more natural pattern of the red shrimp distribution.

8 | Conclusions

- Hierarchical modelling can be a really useful tool for statistical modellers.
- Bayesian is a good option to deal with these models.
- Indeed: “Everybody is a Bayesian. It’s just that some know it, and some don’t ”(T. Raghunathan).
- INLA + SPDE can also be very convenient as they are fast and provides good results (probably not the best ones, but the first ones in your analysis).



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